DO NOW — RECALL

Linked lists using objects

Name: _____ Class: _____ Date: _____

1. What makes a heuristic admissible?

2. What does a partition do to the pivot?

3. What does the head reference identify?

4. Why must a new node's Next be set before replacing an existing Next link?

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

It never overestimates the true cheapest remaining cost.

2. Model answer

Places it at its final index with smaller/equal values to its left.

3. Model answer

The first node in a linked list.

4. Model answer

Otherwise the original successor can become unreachable and part of the list may be lost.

Vocabulary — know these terms

node — An object containing data and a reference to another node.

head — Reference to the first node.

Next — Link to the successor node.

null — No object is referenced.

traversal — Visit nodes successively by following Next references.

NEW LEARNING

Read, annotate and explain

Build a chain

Each Node stores a Value and a Next reference; the final Next is null.

Preserve the successor

Construct 15 with Next set to the old 20 node *before* changing 10.Next.

Finish the insertion

Set 10.Next to 15. Now traversal yields 10→15→20→30.

Explain one step in your own words, using the correct technical terms.

I DO — WORKED EXAMPLE

Follow the worked method

Insert node 7 between 4 and 9 in 4 → 9 → 12.

Predict first

Set newNode.Next to the existing successor 9.

Model the reasoning

Update node4.Next to reference the new 7 node. The 9→12 remainder is preserved.

WE DO — GUIDED PRACTICE

Try → Discuss → Improve

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

For 3→5→8, identify head, 5.Next.Value and 8.Next.

Question 2

Write two C# statements to insert a node with value 12 after node8 in 8→20→30. Give the new traversal.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 3

Describe how to delete node 20 from 10→20→30 if you hold a reference to node10. Explain the complexity of reaching the deletion point and changing the link.

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

Inserting a linked-list node always requires shifting the stored values of later nodes.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.